

A Physics-Informed Deep Learning Approach to Predicting Bilateral Ground Reaction Forces and Centre of Pressure From a Single Forceplate During Gait

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Word Count: 3,000

Submitted to *Gait & Posture* (Special Issue of ICMST 2024)

March 31, 2025

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Abstract

Background:

Measuring bilateral ground reaction forces (GRFs) and centre of pressure (COP) is essential in gait analysis, requiring subjects to step each foot sequentially onto a separate forceplate. However, this requirement often causes multiple trial attempts, especially in patients with neuromusculoskeletal disorders. Consciously targeting the forceplates could also alter walking mechanics, leading to unnatural gait patterns.

Research Question:

This study aimed to (1) develop a novel physics-informed residual recurrent neural network (PI-ResRNN) to predict bilateral GRF and COP during gait using data from a single forceplate and (2) evaluate its accuracy against ground truth obtained across subject groups of different ages and pathologies.

Methods:

Forceplate data from 315 participants, namely healthy participants and patients with six types of neuromusculoskeletal disorders, was collected. Data from 6,765 trials was used to train and validate the PI-ResRNN model to decompose GRF and COP for each foot during the double-contact phase of walking. Model-predicted COP and GRFs were evaluated against the ground truth using root-mean-square errors (RMSE) and relative RMSE (rRMSE), respectively.

Results:

All predicted variables from the PI-ResRNN model demonstrated high consistency with the ground truth, with mean rRMSE values below 0.34%, 0.38%, and 0.56% in the vertical, anteroposterior, and

mediolateral GRFs, respectively, and mean RMSE values for COP below 3.0 mm. The model effectively identified statistical between-group differences compared with the ground truth.

Significance:

The proposed model provides a practical and accurate approach for obtaining bilateral GRF and COP using a single forceplate, benefiting gait analysis in populations with mobility impairments.

Keywords: Ground Reaction Forces; Centre of Pressure; Forceplate; Deep Learning; Physics-Informed Neural Network; Gait Analysis

Highlights

- PI-ResRNN accurately predicts bilateral GRF and COP from a single forceplate
- PI-ResRNN produces accuracy comparable to the precision of forceplates
- Training on large datasets with physics rules ensures biomechanical realism
- PI-ResRNN effectively detects statistically significant group differences
- PI-ResRNN avoids step targeting or repeated trials in clinical gait analysis

Introduction

Accurate measurement of ground reaction forces (GRFs) and centre of pressure (COP) for each foot is essential for analysing foot loading patterns [1-3] and calculating joint moments [4-6]. To achieve this, participants must place each foot on and within a separate forceplate in consecutive steps [7]. This often necessitates multiple practice trials, which can be particularly challenging for individuals with physical weakness or severe neuromuscular or orthopaedic disorders. Repeated trials could also introduce increased fatigue and discomfort for participants, reducing the reliability of gait measurements [7]. On the other hand, consciously targeting the forceplates may induce unnatural gait patterns, substantially impacting GRF measurements [8]. This issue is particularly evident in individuals whose natural stride length is too long (overshooting the forceplate) or too short (landing both feet on a single plate).

Methods have been proposed to decompose the resultant GRF and COP vectors (F and P) during the double-limb support (DLS) measured by a single forceplate to obtain the GRF vectors (F_L and F_T) and the two-dimensional COP position vectors (P_L and P_T) for the leading and trailing foot, respectively [9-14]. However, the core challenge is that this task is inherently an indeterminate problem with ten unknowns but only five measured values [15]. Most existing methods rely on additional assumptions or constraints to obtain a unique solution for bilateral GRF and COP [9-12]. Yet, their prediction accuracy was often much lower than that required for inverse dynamics analysis. Davis and Cavanagh [9] proposed a custom forceplate with four load cells to estimate vertical GRF distribution based on ipsilateral load cell outputs, with errors exceeding 7%. Ballaz *et al.* [10] first introduced a cubic spline transition function to estimate the leading foot's vertical GRF, and others later extended to shear forces and ground reaction moments (GRM) [11, 12]. These approaches reported errors exceeding 2%, 6%, and 9% in vertical, anteroposterior, and mediolateral GRFs, respectively, with COP errors over 13 mm. Such inaccuracies are quite significant because, for example, a 2% GRF error can produce up to 43%

error in the calculated hip joint moments [16]. Moreover, the transition functions are often chosen for mathematical convenience or based on typical gait patterns, with no experimental validation supporting their use.

To avoid predefined transition functions, Bastien *et al.* [14] proposed a linear mapping model to decompose shear GRFs by relating the total anteroposterior GRF to that of the leading foot. It was primarily trained on healthy individuals, with limited validation in pathological populations. Moderate accuracy was reported, with 3.1% and 5.6% errors in the anteroposterior and mediolateral GRF, respectively. They assumed each foot's COP during DLS as a fixed point despite known COP shifts of 20-70 mm during gait and may even greater displacements in patients with neuromuscular or orthopaedic conditions [17]. Note that a 10 mm COP shift can result in a 14% error in calculated peak joint torques [18]. To date, no existing method has demonstrated sufficient accuracy or generalizability to be considered practical for clinical gait analysis. Most methods are validated only in healthy individuals or lack rigorous validation, and none has undergone a thorough accuracy assessment for patient populations. A robust, accurate and validated approach is needed to detect clinically meaningful differences across healthy and pathological populations.

Deep learning offers strong potential for modelling the complex, time-varying relationships between single forceplate and foot-specific GRF and COP by learning directly from real-world data. Recurrent neural networks (RNNs) are well-suited for sequential data, capturing temporal dependencies through feedback connections [19]. Residual RNNs (ResRNNs), which incorporate skip connections, further improve training stability and preserve input-output relationships [20]. However, standard data-driven models may generate outputs that violate biomechanical principles [21]. Physics-informed neural networks (PINNs) address this by embedding physical constraints directly into the loss function, ensuring biomechanically valid predictions [22]. Combining ResRNNs with PINNs bridges data-

driven learning with theoretical biomechanics, making it especially suitable for solving the mathematically indeterminate problem of GRF and COP decomposition.

This study aimed (1) to develop a novel physics-informed ResRNN (PI-ResRNN) model for predicting bilateral GRF and COP during gait using data from a single forceplate and (2) to evaluate the accuracy against the ground truth data and the model performance in assessing the statistical differences between various patient groups and their healthy peers.

Materials and Methods

Subjects

Retrospective gait analysis data from 315 subjects, consisting of 180 healthy participants and 135 patients, from studies that had previously been conducted [23-36] were used in this study (Table 1). There were three age groups: Children, Young, and Older, and six patient groups: Cerebral Palsy (CP), Duchenne Muscular Dystrophy (DMD), Adolescent Idiopathic Scoliosis (AIS), Anterior Cruciate Ligament Deficiency (ACLD), Knee Osteoarthritis (OA), and Stroke (Table 1). All participants underwent screening to confirm the absence of additional musculoskeletal or neurological disorders and to ensure normal or corrected vision during the gait experiment. As no personally identifiable information was involved, and the dataset complied with applicable regulatory and institutional guidelines, the study was exempt from the Institutional Research Board review under the institutional policies.

Experimental Protocol

The gait data of all participants were collected from the same university hospital gait laboratory following a standardised protocol. Each subject walked along a 10-meter walkway equipped with four

forceplates (OR-6-7-1000, AMTI, USA) routinely calibrated *in situ* to ensure measurement accuracy [37]. During the gait trials, participants walked and stepped on the forceplates while focusing on a fixed point ahead without looking at the ground, avoiding intentionally targeting the forceplates [38]. Participants with smaller step lengths achieved natural patterns by walking along the centerline between two adjacent forceplates [39]. Three-dimensional GRF and GRM for each foot were measured at 1200 Hz. Trials were excluded if the participant stepped off the forceplate or between two forceplates. At least 30 successful trials with complete forceplate data for two consecutive steps were obtained for healthy participants, while at least 10 successful trials were obtained for patients.

Forceplate Data Processing

A total of 6,765 trials were obtained for this study. The measured data were processed using a bi-directional 4th-order Butterworth low-pass filter (cut-off: 25 Hz) [40], and the gait events were identified for each foot subsequently [41]. The global coordinate system (GCS) origin was defined on the ground surface, with the X-axis directed to the right, the Y-axis directed anteriorly, and the Z-axis directed vertically. The global GRF and GRM from two consecutive steps from two forceplates were combined to simulate the resultants of the bilateral data on a single forceplate. The COP for each foot was also calculated from the measured data to be used as the ground truth for evaluating model performance in subsequent analyses.

Physics-Informed Residual RNN (PI-ResRNN) Model

A novel PI-ResRNN model was designed to predict GRF and COP for each foot from data of a single forceplate (Figure 1). The model was informed of the physical relationships between the GRF, GRM and COP by incorporating upper bounds into the RNN layers and equality constraints into the loss function. The RNN layers were combined with subtraction connections, which ensured that the GRF, GRM and COP of each foot added up to the total measured data.

The model's architecture consists of an input layer, ten ResRNN layers, and an output layer. The input vector comprised six time-series signals from the GRF and GRM of both feet on a forceplate, namely three force components (anteroposterior, mediolateral and vertical) and 3 moment components. The output vectors consisted of the time series of GRF and COP of the stance phase (DLS and SLS phases) for each foot. Both the input and output layers were normalised by body weight [41]. As the core component of the model, the ten RNN layers extract and process sequential information over time. Each layer consists of 4,096 cells, implemented using bidirectional long short-term memory (LSTM) units [42]. In the first RNN layer, the vertical GRF output passed through a rectified linear unit (ReLU) activation function [43] before being subtracted from the input data through residual subtraction connections. The output data were then passed through the RNN layers and, layer by layer, progressively refined the predicted GRF and GRM of the leading foot. The addition of ReLU activation at each RNN layer ensured that the predictions satisfied the inequality constraint and that the predicted vertical GRF of a foot did not exceed the total measured vertical GRF of both feet, ensuring physically plausible predictions. Following the RNN layers, the output layer generates GRF and GRM for the leading (L) foot. The GRF and GRM for the trailing (T) foot and the COP for each foot were then calculated using the law of mechanical equilibrium.

$$\hat{F}_T = F - \hat{F}_L \quad (1)$$

$$\hat{M}_T = M - \hat{M}_L \quad (2)$$

$$\hat{P}_i = \begin{bmatrix} -\hat{M}_i \cdot Y \\ \hat{F}_i \cdot Z \\ \hat{M}_i \cdot X \\ \hat{F}_i \cdot Z \end{bmatrix}, i \in \{L, T\} \quad (3)$$

where F and M are the total GRF and GRM obtained from a single forceplate; $\hat{F}_i$, $\hat{M}_i$ and $\hat{P}_i$ are the model-predicted GRF, GRM and COP for the leading (L) and trailing (T) foot, respectively; X , Y and Z are the unit vectors of the GCS.

A physics-informed loss function incorporating three commensurable loss terms (i.e., L_M , L_F , and L_P) corresponding to the differences between the experimentally measured and model-predicted GRM, GRF, and COP was used to guide the training of the model. The loss terms were formulated to have the same physical units and avoided the division operations [15], thereby improving numerical stability.

$$L_M = \|\hat{M}_T - M_T\|^2 + \|\hat{M}_L - M_L\|^2 \quad (4)$$

$$L_F = \|P_T \times \hat{F}_T - P_T \times F_T\|^2 + \|P_L \times \hat{F}_L - P_L \times F_L\|^2 \quad (5)$$

$$L_P = \|P_T \times \hat{F}_T - \hat{M}_T\|^2 + \|P_L \times \hat{F}_L - \hat{M}_L\|^2 \quad (6)$$

$$Loss = \lambda_M L_M + \lambda_F L_F + \lambda_P L_P \quad (7)$$

where F_i , M_i and P_i are the experimentally measured GRF, GRM and COP obtained from forceplates of leading (L) and trailing foot (T), respectively; λ_M , λ_F and λ_P are the weighting factors, respectively. The values of weighting factors were determined empirically to be 0.1, 0.1 and 10, respectively.

The PI-ResRNN model was implemented in Python 3.10 using PyTorch [44] and trained with the RTX-4090 GPU. The Adaptive Moment Estimation (Adam) optimiser [45] was used with a learning rate of 0.0001 and exponential decay rates of 0.9 and 0.999 for the first- and second-moment estimates, respectively. The maximum number of training epochs was set to 1000, with a batch size of 256.

Datasets for Model Training, Validation and Testing

The dataset was divided into separate subsets for model training, validation, and testing. To evaluate model accuracy and its ability to detect statistical differences between patient groups and age-matched healthy controls, 15 participants were selected from each patient group, along with 15 participants from each of the three healthy control groups, resulting in a testing set comprising data from 135 participants. The remaining 165 participants were allocated for model training, with an additional 15 participants reserved for validation during the training process. In total, 6,765 trials were available and distributed across the three subsets: 4,305 trials for training, 210 for validation, and 2,250 for testing.

Performance Evaluation

The performance of the PI-ResRNN model was evaluated by its prediction accuracy against the ground truth and for its ability to detect statistically significant differences between patient groups and their age-matched healthy controls. For all trials in the testing dataset, the relative root mean squared errors (rRMSE, %) were calculated for each foot during DLS as the root mean squared error (RMSE) divided by the range of the measured signal [21]. For the COP, RMSE was calculated. For between-group comparisons, the GRF-sharing of the leading limb during DLS was calculated as the time integral of the leading limb GRF divided by the time integral of the total GRF.

Statistical Analysis

For each subject, the rRMSE values for GRF and the RMSE values for COP were averaged across all testing trials. Subsequently, the means and standard deviations of each error metric were calculated across all subjects of each testing subject group. A two-way mixed-design analysis of variance (ANOVA) with one between-subject main factor (group) and one within-subject main factor (limb) was conducted for all calculated error metrics.

To evaluate model performance in group effects detection, independent *t*-tests were used to compare the leading-limb GRF-sharing between each testing patient group and their age-matched healthy controls for the model-predicted and ground truth results, respectively. The corresponding effect sizes and their Pearson correlation coefficients between the model predictions and the ground truth were calculated [21]. The accuracy in detecting statistical between-group differences was also obtained. All statistical analyses were performed using SPSS version 20 (SPSS Inc., Chicago, IL, USA), with the significance level set at 0.05.

Results

For the healthy participants, the PI-ResRNN model showed high consistency with the ground truth in estimating GRFs (Figure 2), with mean rRMSE values of 0.30%, 0.35%, and 0.53% for the vertical, anteroposterior, and mediolateral directions, respectively (Table 2). The model-predicted COP closely aligned with the ground truth (Figure 2), with a mean RMSE of 2.9 mm (Table 2).

In the patient groups, the PI-ResRNN model also showed strong agreement with the ground truth for GRF estimation (Figure 3), with mean rRMSE values of 0.35%, 0.39%, and 0.57% for the vertical, anteroposterior, and mediolateral directions, respectively (Table 2). The predicted COP showed highly consistent with the ground truth (Figure 3), with a mean RMSE of 3.2 mm (Table 2).

No significant differences in prediction errors were observed between groups or limbs for any GRF or COP variables predicted by the PI-ResRNN model (Table 2). Based on the experimental measurements, the Stroke, ACLD, CP, and AIS groups showed significantly increased GRF sharing of the leading limb during DLS compared to their healthy peers. The same results were also predicted by the PI-ResRNN model, with the model-predicted between-group effect sizes showing a strong

correlation ($r=0.99$) with those derived from the ground truth data.

Discussion

The current study developed a novel physics-informed deep-learning approach for predicting bilateral GRF and COP during DLS using data from a single forceplate. By embedding physical laws into the learning architecture and minimising errors across all outputs during training, the PI-ResRNN model learned complex, time-varying patterns directly from data and achieved superior accuracy across all predicted variables compared to existing methods, regardless of age or pathological condition, whether symmetric and markedly asymmetric gait patterns (Table 3). The prediction errors were approaching the measurement precision of commercial forceplates (typically 0.2%-2% of full-scale output). The findings suggest that the model offers a more practical and robust solution for estimating bilateral GRF and COP from a single forceplate, reducing the challenges of multiple trial attempts and the gait alterations caused by weak or patient groups.

The superior performance of the PI-ResRNN model arises from integrating physics-informed constraints within a deep learning framework, effectively combining data-driven modelling with biomechanical principles. The architecture incorporates three key features that enhance predictive accuracy: (1) a physics-informed structure that estimates GRF and GRM for the leading limb, with those for the trailing limb derived via laws of equilibrium; (2) a physics-informed loss function that jointly minimises errors in GRF, GRM, and COP for both limbs; and (3) deep residual connections with ReLU activations to enable stable temporal learning while enforcing physiologically plausible bounds on predicted forces. With these features, the model captures the physical coupling between limbs during DLS, ensuring internal consistency and allowing errors in one limb to be offset by improvements in the other. The combined loss function further reinforces the biomechanical

interdependence among these variables, ensuring numerical precision and physical validity for supporting balanced and interpretable predictions. The deep ResRNN architecture supports hierarchical learning with stable training while enforcing biomechanical constraints, such as non-negative vertical GRFs within the limits of resultant GRF. These mechanisms improve robustness across various gait patterns, including pathological conditions.

The generalizability of the PI-ResRNN model is largely attributed to its training on a large, diverse dataset combined with the representational power of deep learning, enabling high accuracy across heterogeneous populations and the detection of statistical between-group differences. This robustness suggests suitability for clinical use without subject-specific recalibration, making it highly suitable for instrumented gait assessments. In contrast to existing methods, where age-related gait variations may lead to increased errors due to misalignment with predefined transition functions [41], the PI-ResRNN model maintained consistent accuracy. In neuromuscular conditions such as CP, compensatory gait strategies can significantly alter force distributions under both feet, resulting in asymmetrical loading [32] and altered force profiles [30], all of which are difficult to capture using rule-based approaches. The PI-ResRNN model accurately captured disease-specific gait patterns, replicating significant group-level differences in leading-limb GRF sharing with effect sizes closely matching ground truth. Its consistent performance across diverse testing subsets highlights strong robustness and generalizability. These findings support the model's utility in clinical gait analysis, particularly for applications requiring the detection of asymmetries or the monitoring of treatment outcomes across various populations and cases.

This study is the first to introduce a physics-informed, data-driven deep-learning approach for predicting bilateral GRF and COP during DLS using data from a single forceplate. The high accuracy across diverse age groups and clinical populations achieved by the PI-ResRNN suggests its potential in

measuring bilateral GRF and COP in clinical gait analysis while reducing the need for repeated trial attempts or gait modifications. Additional data from a broader range of patient populations will be required to enhance the generalizability and applicability further. Future studies should also examine the impact of decomposed GRF and COP on joint force and moment estimations. Extending this approach to accommodate varied walking conditions, such as slope walking or obstructed gait, could further strengthen its clinical and biomechanical utility.

Conclusions

This study developed a novel physics-informed deep-learning approach for predicting bilateral GRF and COP during DLS using data from a single forceplate. The proposed PI-ResRNN model demonstrated high accuracy across all predicted variables, regardless of age or pathological condition. Notably, the model effectively detected group differences in GRF-related variables. By embedding physical laws into the learning architecture and minimising errors across all outputs during training, the PI-ResRNN was capable of accurately predicting bilateral GRF and COP in individuals with both symmetric and markedly asymmetric gait patterns, including those with hemiplegic cerebral palsy. These findings suggest that the PI-ResRNN outperforms existing approaches based on predefined transition functions or data-driven linear mapping models. Considering its accuracy and generalizability, this approach offers a more practical and robust solution for estimating bilateral GRF and COP from a single forceplate.

TABLES

Table 1. Means (standard deviations) of demographic characteristics of healthy participants, i.e., children (Children: 0–18 years), young adults (Young: 19–64 years), and older adults (Old: ≥ 65 years), and patients with pathological conditions, namely cerebral palsy (CP), adolescent idiopathic scoliosis (AIS), Duchenne muscular dystrophy (DMD), anterior cruciate ligament deficiency (ACLD), stroke, and knee osteoarthritis (OA).

Group	Sample Size (N)	Age (years)	Body mass (kg)	Height (cm)
Healthy participants				
Children	60	9.2 (3.8)	121.1 (28.5)	31.1(17.2)
Young	60	32.3 (12.2)	164.5 (10.6)	67.2 (13.2)
Old	60	72.1 (4.6)	157.3 (7.8)	64.3 (8.2)
Patients with pathological conditions				
CP	60	10.3 (3.4)	132.5 (23.1)	31.0 (13.9)
AIS	15	14.5 (1.9)	152.6 (8.0)	39.2 (7.3)
DMD	15	10.1 (2.7)	126.3 (15.4)	32.4 (14.0)
ACLD	15	35.5 (13.4)	167.4 (5.7)	62.8 (8.4)
Stroke	15	69.9 (3.9)	161.2 (7.4)	62.1 (9.6)
Knee OA	15	71.4 (4.6)	155.6 (4.5)	63.7 (8.2)

Table 2. Means (standard deviations) of PI-ResRNN model prediction errors in bilateral GRF and COP for healthy and patient groups. Errors are presented as relative root mean square error (rRMSE) for GRF and root mean square error (RMSE) for COP.

Variables	Group	Error		p-value	
		Leading Limb	Trailing Limb	Limb effect P _L	Group effect P _G
Vertical GRF (%)	Children	0.26 (0.11)	0.27 (0.11)	0.79	0.11
	Young	0.32 (0.16)	0.33 (0.21)		
	Old	0.35 (0.19)	0.33 (0.17)		
	CP	0.35 (0.21)	0.35 (0.23)		
	AIS	0.41 (0.34)	0.38 (0.25)		
	DMD	0.39 (0.27)	0.37 (0.25)		
	ACLD	0.34 (0.20)	0.35 (0.21)		
	OA	0.28 (0.19)	0.27 (0.21)		
	Stroke	0.42 (0.19)	0.40 (0.22)		
Anteroposterior GRF (%)	Children	0.31 (0.15)	0.32 (0.14)	0.87	0.11
	Young	0.39 (0.20)	0.39 (0.23)		
	Old	0.36 (0.20)	0.33 (0.16)		
	CP	0.36 (0.19)	0.37 (0.16)		
	AIS	0.33 (0.15)	0.34 (0.17)		
	DMD	0.52 (0.34)	0.43 (0.26)		
	ACLD	0.43 (0.25)	0.46 (0.28)		
	OA	0.35 (0.25)	0.36 (0.26)		
	Stroke	0.38 (0.27)	0.40 (0.27)		
Mediolateral GRF (%)	Children	0.56 (0.25)	0.57 (0.25)	0.50	0.09
	Young	0.56 (0.22)	0.61 (0.26)		
	Old	0.47 (0.27)	0.47 (0.25)		
	CP	0.67 (0.39)	0.73 (0.42)		
	AIS	0.67 (0.39)	0.64 (0.39)		
	DMD	0.52 (0.43)	0.58 (0.49)		
	ACLD	0.50 (0.39)	0.58 (0.55)		
	OA	0.58 (0.36)	0.62 (0.40)		
	Stroke	0.41 (0.39)	0.40 (0.36)		
COP (mm)	Children	2.88 (3.04)	2.85 (2.16)	0.97	0.41
	Young	2.72 (1.12)	2.68 (1.06)		
	Old	3.08 (2.40)	3.06 (2.28)		
	CP	2.69 (1.79)	2.52 (1.60)		
	AIS	3.48 (2.01)	3.55 (2.20)		
	DMD	3.17 (1.73)	3.08 (1.59)		
	ACLD	3.12 (1.60)	3.35 (1.57)		
	OA	3.46 (1.84)	3.60 (1.99)		
	Stroke	2.67 (1.43)	2.66 (1.46)		

Table 3. Mean errors for GRF and COP between the proposed PI-ResRNN method and existing methods for healthy participants and patients.

Methods	Number of subjects	Variable			COP (mm)
		Vertical GRF (%)	Anteroposterior GRF (%)	Mediolateral GRF (%)	
Assessment using healthy participant data					
PI-ResRNN Model	180	0.3	0.4	0.5	2.9
Davis & Cavanagh [9]	1	7.7	-	-	-
Ballaz <i>et al.</i> [10]	6	3.8	-	-	-
Villeger <i>et al.</i> [11]	7	3.8	6.0	18.0	13.5
Samadi <i>et al.</i> [12]	30	2.9	6.3	9.5	-
Meurisse <i>et al.</i> [13]	27	1.8	-	-	-
Bastien <i>et al.</i> [14]	27	-	3.1	5.6	-
Assessment using patient data					
PI-ResRNN Model	135	0.4	0.4	0.6	3.2
Meurisse <i>et al.</i> [13]	88	2.5	-	-	-
Bastien <i>et al.</i> [14]	88	-	9.8	12.1	-

FIGURE CAPTIONS

Figure 1: Architecture of the physics-informed residual recurrent neural network (PI-ResRNN) model for decomposing bilateral ground reaction forces (GRF) and centre of pressure (COP) from data obtained using a single forceplate. The model receives six time-series signals (three force and three moment components) as input and passes them through ten residual bidirectional long short-term memory (LSTM) layers, each containing 4,096 cells. Rectified linear unit (ReLU) activations and residual subtraction connections are incorporated to enforce physical constraints. The output layer produces GRF and ground reaction moment (GRM) for the leading foot, while the GRF and GRM for the trailing foot and the COP for both feet are computed based on the laws of mechanical equilibrium. A custom physics-informed loss function is used during training to ensure consistency between the model predictions and measured data in terms of force, moment, and pressure.

Figure 2: Model-predicted and experimentally measured GRFs and COP during gait for both the leading (blue) and trailing (red) limbs in a healthy young adult (age: 24 years, height: 179 cm, weight: 68 kg). (A–C) show the vertical, anteroposterior, and mediolateral components of GRF, respectively, plotted over normalised stance time (%). (D) illustrates the COP trajectories in the transverse plane, with mediolateral position on the x-axis and anteroposterior position on the y-axis. The combined data from a single forceplate, which served as the model input, are shown as solid black lines. The model-predicted outputs are shown as dashed lines, and the gold-standard measurements from two separate forceplates are shown as solid grey lines.

Figure 3: Model-predicted and experimentally measured GRFs and COP during gait for both the leading (blue) and trailing (red) limbs in a child with hemiplegic cerebral palsy (age: 8 years, height: 120 cm, weight: 17 kg). (A–C) show the vertical, anteroposterior, and mediolateral components of GRF, respectively, plotted over normalised stance time (%). (D) illustrates the COP trajectories in the

transverse plane, with the x-axis indicating the mediolateral position and the y-axis indicating the anteroposterior position. The combined data from a single forceplate, which served as the model input, are shown as solid black lines. The model-predicted outputs are shown as dashed lines, and the gold-standard measurements from two forceplates are shown as solid grey lines.

Figure 1

Loss Term for Bilateral GRF $L_F = \ \mathbf{P}_T \times \widehat{\mathbf{F}}_T - \mathbf{P}_T \times \mathbf{F}_T\ ^2 + \ \mathbf{P}_L \times \widehat{\mathbf{F}}_L - \mathbf{P}_L \times \mathbf{F}_L\ ^2$
Loss Term for Bilateral COP $L_P = \ \mathbf{P}_T \times \widehat{\mathbf{F}}_T - \widehat{\mathbf{M}}_T\ ^2 + \ \mathbf{P}_L \times \widehat{\mathbf{F}}_L - \widehat{\mathbf{M}}_L\ ^2$

Physics-Informed Loss Function via Def

Figure 2

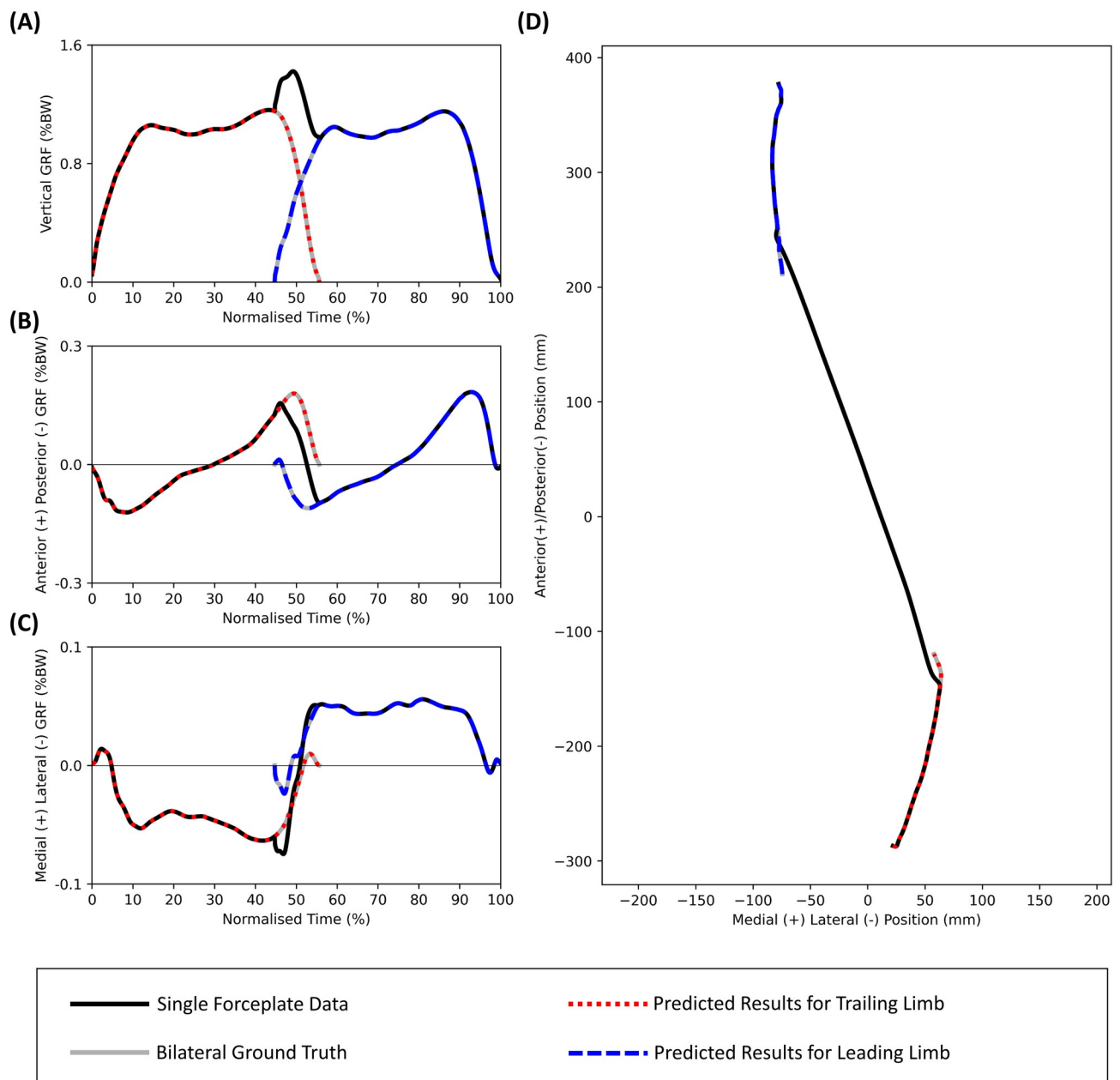
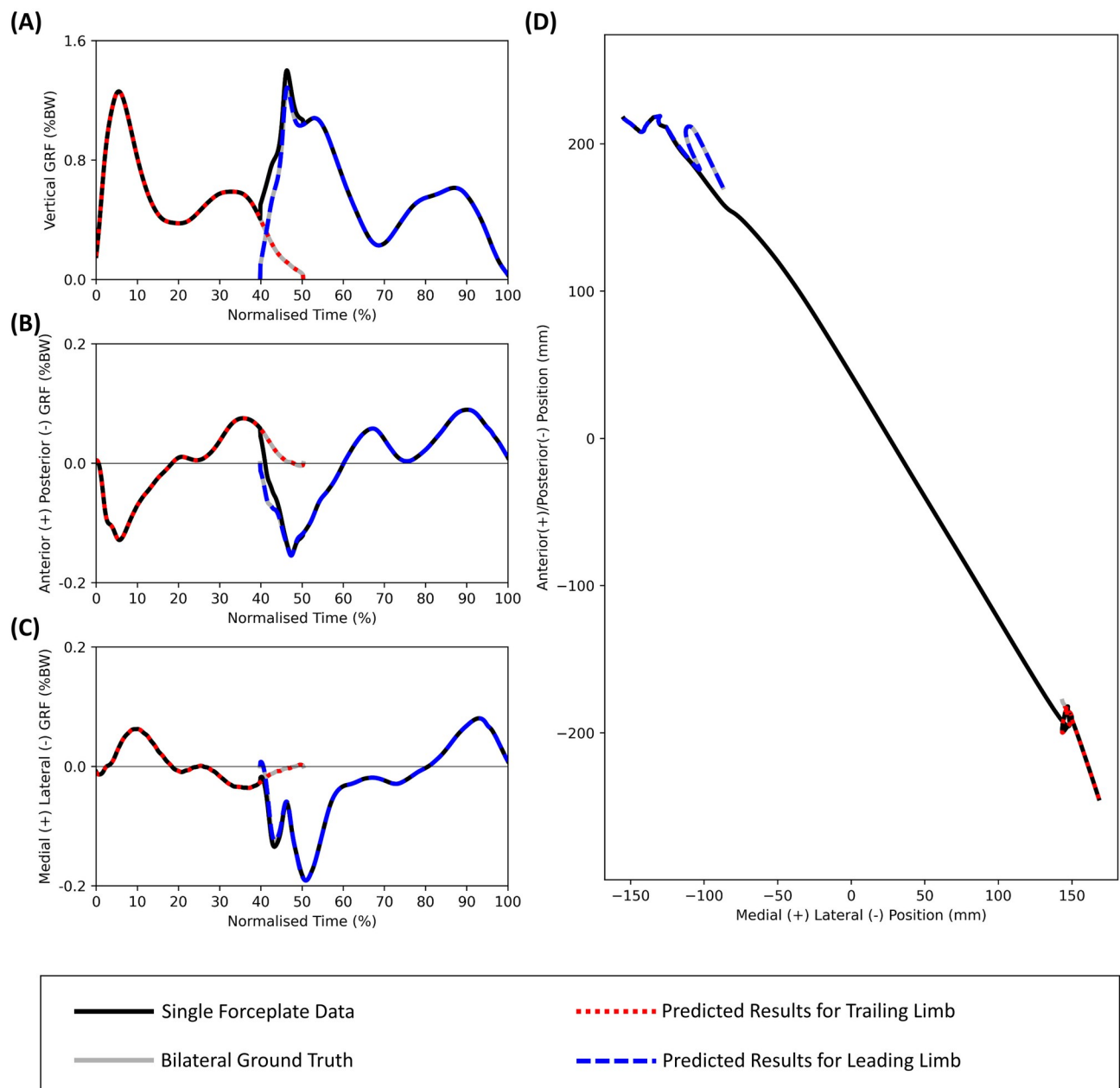


Figure 3



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